

# Basem Mohamed

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## EDUCATION

### University of Houston

*Bachelor of Science, Computer Science*

**Aug 2023 – May 2027**

*Houston, TX*

## PROJECTS

### **AI Room Remodeler (Hackathon Winner)** | *Express.js, TailwindCSS, SerpAPI, Gemini* **Oct 2025**

- Earned **2nd** Place by CodeRED and Best Use of Gemini API by Major League Hacking.
- Built a full-stack prototype that turns a room photo into redesign options plus a shoppable furniture list with live product sourcing.
- Integrated Google Gemini 2.5 Flash-Image & Pro models for image-to-image generation and JSON-based furniture metadata extraction.
- Implemented SerpAPI's Google Shopping Scraper to turn Gemini outputs into product listing links.

### **Dream Maker (Image Generator with Daily Quotas)** | *React, Node.js, SQLite*, **May – Sept 2025**

- Built a React web app that turns user provided descriptions of dreams into images using a React/Vite front end and an Express API.
- Added cost/abuse controls by issuing cookie-based anonymous user IDs, enforcing **20** generations per **24** hours, and deploying SQLite-backed persistent usage tracking with a UI countdown.
- Improved reliability and API hygiene with server-side validation, clear HTTP errors (400/403/500), and atomic quota updates in SQLite to prevent counter desync.

### **The Scholar's Companion** | *Flask, Google Search API, Bootstrap, C++* **May 2023 – June 2024**

- Developed a tool that ranks Google results by reading difficulty for non-native English speakers.
- Designed a readability-scoring algorithm grounded in established linguistic research to classify educational text from grade-school through college level.
- Collaborated in a two-person team to deploy and maintain the application for over **12** months, supporting ~**50** monthly active users and resolving user-reported issues.

## WORK EXPERIENCE

### **Freelance Fullstack Developer**

*E-commerce Freight & Shipping System / WooCommerce, FedEx API*

**Feb 2026 – Present**

*Houston, TX*

- Designed a hybrid shipping system routing orders between live FedEx rates and a custom freight workflow, ensuring accurate pricing for all products
- Eliminated manual shipping operations by automating rate calculation based on customer address and order details, and replacing freight handling with a scalable, on-demand quote system
- Redesigned an e-commerce storefront to reduce purchase friction, contributing to a reported **13x** increase in visitor-to-buyer conversion through stronger visual hierarchy and CTAs.

### **Multi-Level Math Instructor**

*Mathnasium*

**Oct. 2023 – Sept. 2024**

*Houston, TX*

- Taught and coached 40+ students across multiple grade levels by translating complex ideas into simple steps and adjusting explanations in real time.
- Diagnosed learning gaps through short assessments and observation and created targeted practice plans that improved student confidence and consistency.

## TECHNICAL SKILLS

**Languages:** Java, Python, C/C++, JavaScript, HTML, CSS

**Frameworks:** React, Node.js, Express, Flask, Vite

**Technologies:** PostgreSQL, UX Best Practices, Data Structures, Algorithms, Software Development

**Developer Tools:** Git, VS Code, ESLint

**Spoken Languages:** English (Fluent), Arabic (Fluent)